NOURELDIN OSAMA FATHY ELANANY

AI ENGINEER | PYTHON | LARGE LANGUAGE MODELS (LLMs) | HEALTHCARE RAG | COMPUTER VISION

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PROFESSIONAL SUMMARY

Results-driven Artificial Intelligence Graduate (Faculty of Artificial Intelligence, Benha University — Class of 2026) with specialized expertise in Python, Large Language Models (LLMs), Healthcare Retrieval-Augmented Generation (RAG) architectures, and Computer Vision (Score: 98% Excellent from National Telecommunication Institute — NTI). Certified by NVIDIA across Deep Learning, LLMs, and Prompt Engineering competencies. Practical engineering experience at NOVEGATES and NamaSoft developing end-to-end AI pipelines, text and sentiment classification models (BiLSTM), dense vector retrieval (ChromaDB, FAISS, Milvus), and dynamic query routing (LangGraph). Proven track record optimizing model inference latency, reducing LLM token costs by up to 70%, and eliminating hallucinations through prompt-grounded context.

TECHNICAL SKILLS

- **Programming & Languages:** Python (OOP, Clean Architecture, AsyncIO), SQL, Java (LangChain4j enterprise integration).
- **LLMs & Generative AI:** Retrieval-Augmented Generation (RAG), Prompt Engineering, Dynamic Routing (LangGraph), Semantic Caching, Hallucination Reduction, OpenRouter API, Chunking & Tokenization.
- **Vector Databases & Search:** ChromaDB, FAISS, Milvus, Zilliz, Dense Embeddings (Sentence Transformers), HNSW Indexing, Semantic Similarity Retrieval.
- **Deep Learning & NLP:** PyTorch, TensorFlow, Keras, scikit-learn, BiLSTM, LSTM, CNN-LSTM, Text Classification, Sentiment Analysis, TF-IDF.
- **Computer Vision (98% NTI Distinction):** OpenCV, MediaPipe (21 3D Hand Landmarks, 30+ FPS), Multi-Modal Medical Image Fusion (94% Accuracy), Object Detection, Image Preprocessing.
- **Deployment & Tools:** Gradio, Streamlit, Flask REST APIs, Docker, Git, GitHub, Linux / CLI, Agile Methodologies.

PROFESSIONAL EXPERIENCE

AI Data & Pipelines Specialist — NOVEGATES
Jun 2026 – Oct 2026

- Spearheaded domain-specific data collection, validation, and preprocessing pipelines for healthcare AI applications.
- Engineered automated data-cleansing routines ensuring high dataset integrity for downstream NLP classification models.
- Collaborated within a cross-functional engineering team utilizing version control (Git) and agile development workflows.

AI Engineering Trainee — NamaSoft Jul 2025 – Nov 2025

- Completed an intensive 4-month professional AI Engineering program focusing on end-to-end Python ML and DL pipelines.
- Implemented and benchmarked neural architectures (BiLSTM, CNN-LSTM) for sentiment analysis and sequential text classification.
- Built lightweight, modular application interfaces using Streamlit and Flask to demonstrate real-time model predictions.

Computer Vision Specialist Trainee (Score: 98% Distinction) — NTI & ITIDA Jul 2025 – Aug 2025

- Completed 120-hour intensive Computer Vision program achieving the highest distinction evaluation of 98% (Excellent).
- Implemented real-time video processing pipelines utilizing OpenCV and MediaPipe; optimized inference models for edge latency.

SELECTED AI & ENGINEERING PROJECTS

WebMD AI Pipeline: ML, BiLSTM NLP & Medical RAG End-to-End Healthcare AI

Python, scikit-learn, BiLSTM, ChromaDB, OpenRouter API, Streamlit | github.com/NourElanany

- Built an end-to-end healthcare review intelligence pipeline spanning automated data cleaning, drug-effectiveness classification, and review sentiment analysis using word- and character-level TF-IDF features with a BiLSTM network.
- Implemented vector retrieval with ChromaDB and dynamic condition filters to power prompt-grounded, hallucination-free medical summaries via OpenRouter LLM API.

Enterprise Java RAG System with LangChain4j & Milvus Sub-15ms Latency

LangChain4j, Milvus, Zilliz, Java, Vector Search | github.com/NourElanany

- Designed and implemented enterprise-grade RAG pipelines utilizing LangChain4j integrated with MilvusEmbeddingStore.
- Applied HNSW vector indexing algorithms achieving sub-15ms dense retrieval latency across technical repositories.

Dynamic LLM Routing Gateway & Cost Optimizer -70% Token Cost

Python, LangGraph, Sentence Transformers, Streamlit | github.com/NourElanany

- Engineered an intelligent query router using LangGraph that evaluates prompt complexity dynamically to route queries between lightweight vs. frontier models, slashing token costs by up to 70%.
- Incorporated semantic caching with Sentence Transformers, retry/fallback logic, and built a dual-mode Streamlit testing interface.

Clinical Drug-Review Semantic Search & Zero-Hallucination RAG Zero Hallucination

Python, Gradio, Sentence Transformers, FAISS, Prompt Engineering | github.com/NourElanany

- Developed an interactive Gradio application indexing healthcare reviews; engineered semantic chunking, metadata-aware vector embeddings, and top-k retrieval using FAISS.
- Eliminated clinical hallucinations by constraining response generation strictly to verified retrieved drug review passages.

Real-Time Video Computer Vision & Hand Landmark Tracking — 30+ FPS Live

Python, OpenCV, MediaPipe, NumPy, Real-Time Streaming | **github.com/NourElanany**

- Engineered high-throughput computer vision pipeline tracking 21 3D hand landmarks in live video streams at 30+ FPS.
- Integrated real-time gesture telemetry, bounding-box calculation, and low-latency rendering for interactive applications.

Multi-Modal Medical Image Fusion Diagnostic System — 94% Accuracy

Python, PyTorch, OpenCV, Scikit-image, Deep Learning | **github.com/NourElanany**

- Implemented dual-modality radiological image fusion combining structural MRI and functional PET scans, achieving a 94% diagnostic classification accuracy to assist clinical decision-making.

EDUCATION & ACADEMIC HONORS

Bachelor of Science in Artificial Intelligence — Benha University — 2022 – 2026 (Graduated)

- Faculty of Artificial Intelligence, Benha University — Major: AI Engineering & Data Science (Class of 2026).
- National Telecommunication Institute (NTI) Professional Program: Graduated with 98% (Excellent Distinction).

CERTIFICATIONS & VERIFIED CREDENTIALS

- **NVIDIA:** Rapid Application Development with Large Language Models (LLMs) — Credential ID: `szVIDKH8Qty0uNZQ-XoS1w` (Aug 2025)
- **NVIDIA:** Getting Started with Deep Learning — Credential ID: `etKar9diTCytwki9TmGvgA` (Aug 2025)
- **NVIDIA:** Building LLM Applications With Prompt Engineering — Credential ID: `xdIqRIVRSx60akOMIOFFMQ` (Aug 2025)
- **NTI & ITIDA:** Computer Vision Professional Training (Score: 98% Distinction) — Student ID: `230351` (120 Hours, Aug 2025)
- **Certificate of Achievement:** Generative AI & Applied LLMs Applied Engineering (90 Hours, 3 Production RAG Projects)
- **Udemy & SDE Arts:** The Python Programming A-Z Definitive Diploma — Credential ID: `UC-b1d5e03e-68ce-4939-ae7f-3b976c3ebf00`
- **Udemy & MTF Institute:** Python Development and Python Programming Fundamentals — Credential ID: `UC-3d38dcf0-860e-4d1b-9cb7-d599c09af47b`

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency